

Question

Sparingly soluble salt **A** decomposes upon heating to produce four substances **B, C, D, E** (reaction 1). **B** is a black solid, insoluble in dilute hydrochloric acid and dilute sulfuric acid, but soluble in sodium sulfide solution; **B** is also soluble in nitric acid, accompanied by the formation of **F**; **B** reacts with a certain non-metallic element upon heating to 1000°C, and cooling yields **C**. **C** is a liquid at room temperature; **C** reacts with chlorine gas at high temperature to produce a relatively common organic solvent **G**, with byproduct **H** also obtained (reaction 2), **H** is an orange-yellow liquid at room temperature; **H** can continue to react with **C** to obtain **G** (reaction 3). **D** is a gas at room temperature and can be prepared by reacting a highly toxic potassium salt with copper sulfate solution (reaction 4); upon adding ferric chloride solution to the resulting suspension, gas **D** is still generated (reaction 5); **D** burns in an appropriate amount of pure oxygen to produce CO and **E**; **E** is a gas at room temperature with a density slightly less than that of air.

Reactions 1-3 are chemical equations, reactions 4 and 5 are ionic equations, all balanced with the simplest stoichiometric coefficients. The following statements are made:

1. The chemical formula of **A** has an even number of atoms.
2. In the chemical equation for reaction 1, the sum of the coefficients of all species is a prime number.
3. The number of binary compounds among **A** — **H** is equal to 4.
4. In the ionic equations for reactions 4 and 5, the sum of the reactant coefficients are equal.
5. The sum of the coefficients of all species in the chemical equation for reaction 2 is less than the sum of the product coefficients in the chemical equation for reaction 3.

The option that contains all the correct statements is:

A. All other options are incorrect

B. 1, 2

C. 1, 3

D. 1, 4

E. 1, 5

F. 2, 3

G. 2, 4

H. 3, 4

I. 3, 5

J. 2, 5

K. 4, 5

L. 1, 2, 4

M. 1, 3, 4

N. 1, 3, 5

O. 2, 3, 4

P. 2, 3, 5

Q. 2, 4, 5

R. 3, 4, 5

S. 1, 2, 3, 4

T. 1, 2, 3, 5

U. 2, 3, 4, 5

V. 1, 2, 4, 5

W. 1, 2, 3

X. 1, 4, 5

Y. 1, 2, 3, 4, 5

Answer

Correct Answer: **K**

Detailed Explanation

The entry point of this question lies in the preparation of **D**. Highly toxic potassium salts are generally considered to be KCN, then **D** is the gas (CN)₂. Since a suspension is formed, CuCN precipitate is produced. Therefore, the ionic equation for reaction 4 is:

CHECKPOINT

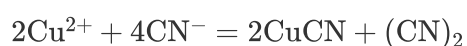
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Highly toxic potassium salts are generally considered to be KCN

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D is the gas (CN)₂

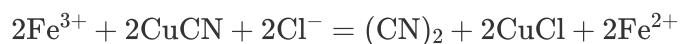


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Reaction 4: $2\text{Cu}^{2+} + 4\text{CN}^{-} = 2\text{CuCN} + (\text{CN})_2$

Upon adding ferric chloride solution, the gas $(\text{CN})_2$ is still generated, which is obviously an oxidation reaction of $\text{Fe}(\text{III})$; therefore, the equation for reaction 5 is


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For the ionic equations of reactions 4 and 5, the sum of the reactant coefficients is 6, so statement 4 is correct.

$(\text{CN})_2$ combustion produces CO and a gas with a density slightly less than air, so its relative molecular mass is less than 29, which is obviously N_2 . Therefore, **E** is N_2 .

CHECKPOINT

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E is N_2

A decomposes to produce N_2 and $(\text{CN})_2$, so it is obviously a thiocyanate; **B** is insoluble in acid but soluble in sodium sulfide solution, which is obviously a metal sulfide. Common metal thiocyanates with black sulfides are considered to be Cu , Hg ; however, CuS is insoluble in nitric acid, so **A** is $\text{Hg}(\text{SCN})_2$ and **B** is HgS .

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A decomposes to produce N_2 and $(\text{CN})_2$, which is a thiocyanate

CHECKPOINT

1 PTS

CuS is insoluble in nitric acid, therefore **A** is $\text{Hg}(\text{SCN})_2$

CHECKPOINT

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B is HgS

HgS dissolves in nitric acid to produce **F**, and **F** reacts with a non-metallic element to obtain liquid **C**, which can react with chlorine to produce an organic solvent. Therefore, the non-metallic element must be carbon; thus, the organic solvent **G** is likely to be CCl_4 . Considering that **F** contains sulfur, it can be assumed that the liquid **C** is CS_2 , and the orange-yellow liquid **H** is S_2Cl_2 . **F** should be elemental S.

CHECKPOINT

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Organic solvent **G** is CCl_4

CHECKPOINT

1 PTS

Orange-yellow liquid **H** is S_2Cl_2

CHECKPOINT

1 PTS

Liquid **C** is **CS₂**

CHECKPOINT

1 PTS

F should be S

According to the chemical formulas, there are 5 binary compounds among **A** – **H**, so statement 3 is incorrect. **Hg(SCN)₂** contains 7 atoms, so statement 1 is incorrect.

Hg(SCN)₂ decomposes to produce **CS₂**, **HgS**, **N₂** and **(CN)₂**; writing reaction 1:



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Reaction 1: $4\text{Hg}(\text{SCN})_2 = 4\text{HgS} + 2\text{CS}_2 + 3(\text{CN})_2 + \text{N}_2$

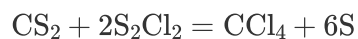
The sum of the equation coefficients is 14, which is not a prime number, so statement 2 is incorrect.

Reactions 2 and 3 can be written according to the chemical formulas:

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Reaction 2: $\text{CS}_2 + 3\text{Cl}_2 = \text{CCl}_4 + \text{S}_2\text{Cl}_2$



CHECKPOINT

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Reaction 3: $\text{CS}_2 + 2\text{S}_2\text{Cl}_2 = \text{CCl}_4 + 6\text{S}$

According to the coefficients, statement 5 is correct.

Therefore, statements 4 and 5 are correct, and option K is correct.